Overnight Price Discovery and the Internationalization of a Currency: The Case of the Korean Won

Jianxin Wang

March 2014

Abstract

This paper proposes a measure for the level of internationalization of a currency. It is based on the relative share of price discovery, termed the information share, in overnight offshore markets. I show that the overnight offshore markets play an increasingly important role in determining the KRW/USD exchange rate. The information share of the overnight offshore markets has increased from less than 10% in 1999-2000 to above 50% in 2011-2012. The variations in the overnight information share are consistent with variations in overnight market efficiency and investor learning.

Keywords: Currency internationalization, overnight price discovery, information share, offshore markets.

JEL Classification: F31, F36, G15

Jianxin Wang (Jianxin.wang@uts.edu.au) is from UTS Business School, University of Technology Sydney, Sydney Australia 2007. Research support from the Australian Research Council and the Commonwealth Bank of Australia is gratefully acknowledged. All errors are my own.

Electronic copy available at: https://ssrn.com/abstract=1972641
Electronic copy available at: http://ssrn.com/abstract=1972641

I. Introduction

The internationalization of a currency has many different dimensions. The currency must be freely traded against other currencies, is widely used for international transactions in goods and services, and is widely accepted as the denominating currency for contracts and assets, e.g. bank accounts and bonds. According to McCauley (2006), "a telling sign of internationalisation is a non-resident issuer of a bond denominated in the domestic currency that is sold offshore to non-resident investors." This multi-dimension nature makes it very difficult to measure the level of internationalization of a currency. For any currency, one can measure its level of internationalization by the percentage of international transactions denominated in the currency, or the onshore versus offshore trading of assets denominated in the currency, or simply the offshore trading of the currency itself among non-residents. Each of these measures captures a different aspect of the internationalization. Take the Australian dollar (AUD) as an example. During 2004-05, 59% of AUD-related currency trading occurred offshore; 30% of AUD-denominated bonds were issued offshore; 78% of offshore AUD-denominated bonds were issued by non-Australian borrowers; 15% of onshore AUD-denominated bonds were issued by non-Australian borrowers; McCauley (2006, tables 1 and 2). While AUD is regarded as highly internationalized, it is hard to take any one of these numbers as a measure for AUD internationalization. To facilitate cross-currency comparisons and to measure the impact of relevant policies, an aggregate measure is needed for the internationalization of a currency.

This paper proposes such a measure of currency internationalization based on the contribution of overnight offshore markets to the price discovery of the currency. Overnight is defined as the period outside the local business hours. Overnight price discovery reflects the aggregate impact of overnight news and events, offshore holding and trading of financial assets/instruments denominated in the currency, and broadly the economic and financial

Electronic copy available at: https://ssrn.com/abstract=1972641
Electronic copy available at: http://ssrn.com/abstract=1972641

integration with the global economy. In this paper, price discovery is measured in two ways: the information share (IS) based on the structural vector autoregressive (VAR) model of Wang and Yang (2011) and a non-parametric measure, the weighted price contribution (WPC). Wang and Yang (2011) modifies the seminal Hasbrouck (1995) model to measure price discovery in sequential markets without overlapping trading time. It is ideally suited for comparing overnight and daytime price discovery. The WPC was proposed by Barclay and Warner (1993) and was first adopted as a price discovery measure by Cao, et al. (2000). Here I provide a brief comparison between the IS and the WPC. Wang and Yang (2012) offers in-depth theoretical and empirical comparisons. In addition to the measures of overnight price discovery, I use the unbiasedness regression of Biais, Hillion, and Spatt (1999) to estimate the impact of overnight price movements on daily returns. Greater overnight price discovery indicates greater information content in overnight price movements, which should lead to greater impact on daytime prices and returns.

The proposed measures are used to evaluate the progress in the internationalization of the Korean won (KRW). The Korean government has engaged in an effort to internationalize the KRW for over a decade; see Rhee (2009) and Kim and Suh (2009). In the early 1990s, KRW was pegged to the U.S. dollar and its value was largely determined by the Bank of Korea. It was relatively insular with no offshore spot or derivative trading. With the deepening of the market liberalization, there are more foreign investments in Korea, driving up the demand for hedging the KRW exchange rate risk. The offshore trading of the non-deliverable forwards (NDF) has flourished. The daily NDF trading increased from US$0.5~1 billion in 2003 (Ma, et al., 2004), to US$2.6 billion in 2005 and US$9.4 billion in 2008 (Rhee, 2009). Table 1 shows the global trading of KRW, including spot and derivatives, as reported in the BIS triennial surveys of the foreign exchange market activity. The estimated offshore trading is based on the method of McCauley (2006) and Kim and Suh (2009). While offshore

Electronic copy available at: https://ssrn.com/abstract=1972641

spot trading of KRW is still prohibited, offshore trading of KRW derivatives has increased from 14% of global KRW trading in 2001 to 37% in 2010. Active offshore trading allows timely information to be incorporated into the value of KRW therefore enhances price discovery in overnight offshore markets. If the KRW is increasingly internationalized with greater offshore holding of KRW-denominated assets and greater offshore trading KRW-related financial products, one would expect to see the overnight IS and WPC rising over time. Overnight returns should have greater information content and greater explanatory power for daily returns. [TABLE 1 ABOUT HERE]

My analyses are based on the spot opening and closing values of the KRW/USD exchange rate during Korean business hours as reported by the Bank of Korea. The sample spans fourteen years from January 1, 1999 to December 31, 2012. I avoid earlier periods when KRW was under a daily trading band or affected by the Asian financial crisis. The main findings are the following:

- There is a strong divergence of opinions between the daytime onshore markets and the overnight offshore markets. The daytime and overnight returns are negatively correlated in every year in the sample. At the onset of the global financial crisis in 2008, the onshore investors were more pessimistic than offshore investors: KRW depreciated a total of 32% during daytime trading but appreciated 2.8% in overnight trading. The sentiment reversed in 2010: KRW appreciated 18.6% during daytime trading but depreciated 16% in overnight offshore markets.
- There is a strong upward trend in overnight price discovery over the sample period. The overnight information share rose from less than 10% in 1999-2000 to over 50% in 2011-12. The overnight WPC rose from less than 20% in 1999-2000 to around 60% in 2011-12. Since there is no offshore spot trading of KRW, the evidence is consistent with the rising impact of the offshore information flow and derivatives trading.

Electronic copy available at: https://ssrn.com/abstract=1972641

- There is evidence of daytime investors learning from overnight price movements. Overnight returns have greater explanatory power for daytime returns in recent years. The low coefficients of the unbiasedness regression in 2005 and 2009 coincide with low overnight IS and WPC in these periods. The R^2 of the unbiasedness regression has a very similar time trend to those of the IS and the WPC.
- Information flow over a calendar week is heavily concentrated on weekdays. The per day information share on weekends is 4.25% compared to 20.5% on weekdays. The weekend information share has not increased over time. These findings further validates the IS and the WPC as measures for offshore information flow and for currency internationalization.

While the price discovery of KRW is increasingly internationalized, it is much lower than currencies such as the Australian dollar (AUD) or the Japanese yen (JPY). Wang and Yang (2011) estimate that around 70% of AUD and JPY price discovery occurs outside Asian business hours. Taking into account currency trading in Hong Kong and Singapore, price discovery outside the home markets for AUD and JPY is likely to be much higher than 70%. Therefore the KRW internationalization has a long way ahead compared to AUD and JPY.

The paper is organized as follows: Section II explains the two price discovery measures and their relationship. Section III first discusses the sample construction and the summary statistics. It then presents the estimated overnight WPC and IS for the KRW/USD exchange rate. The variations in the WPC and IS are linked to the unbiasedness regression. Section IV offers a brief summary.

II. Measuring Pricing Discovery

This section presents two popular measures of price discovery: the WPC first used by Cao, et al. (2000) for measuring price discovery, and the IS of Wang and Yang (2011) which is an extension of the IS measure of Hasbrouck (1995). I also discuss briefly the relationship between the two measures.

Electronic copy available at: https://ssrn.com/abstract=1972641

1. The Weighted Price Contribution

The WPC was first proposed by Barclay and Warner (1993) to measure relative price movements associated with different transaction sizes. It was later used to measure price discovery during the pre-opening period (Cao, et al. 2000), across trading venues (Huang, 2002), during overnight trading (Barclay and Hendershott, 2003, 2008), during opening and closing call auctions (Ellul, et al. 2005), and across dealers in different geographic locations (Anand, et al. 2010).

Here I provide a brief description of the WPC. It is a non-parametric method that calculates the weighted average return attributed to a market or time period. In this case, I want to compare overnight price changes against daytime price changes. Unlike most central banks who report their exchange rates at the end of the local business day, the BOK reports the opening, the highest, the lowest, and the closing values of the KRW/USD exchange rate during the Korean business hours. Let $p_{O,t}$ and $p_{C,t}$ be the opening and closing prices of the KRW/USD exchange rate on day t. The overnight return is $r_{N,t} = \ln(p_{O,t}/p_{C,t-1})$ and the daytime return is $r_{D,t} = \ln(p_{C,t}/p_{O,t})$. The daily return is $r_t = r_{N,t} + r_{D,t} = \ln(p_{C,t}/p_{C,t-1})$. Over the sample period of T days, the overnight WPC is defined as

$$(1) \qquad WPC_N = \sum_{t=1}^{T} \frac{r_{N,t}}{r_t}\left(\frac{|r_t|}{\sum_{\tau=1}^{T}|r_\tau|}\right) = \sum_{t=1}^{T} \frac{r_{N,t}}{r_t} w_t$$

So WPC_N is the weighted average ratio $\frac{r_{N,t}}{r_t}$. The weight $w_t = \frac{|r_t|}{\sum_{\tau=1}^{T}|r_\tau|}$ deals with the case where r_t is near or equal to zero. The daytime WPC, WPC_D, is defined similarly, with $r_{D,t}$ replacing $r_{N,t}$ on the right-hand side of (1). Clearly $WPC_N + WPC_D = 1$ as $r_t = r_{N,t} + r_{D,t}$.

2. Information Share

Hasbrouck (1995) first proposed the information share that captures the relative changes in the efficient price across parallel markets where trading of the same asset takes place simultaneously. Wang and Yang (2011) extend the model of Hasbrouck (1995) to the

Electronic copy available at: https://ssrn.com/abstract=1972641

case of non-overlapping time periods or markets. Using the above notations, the daily return vector, $R_t = [r_{N,t}, r_{D,t}]'$, is modelled as a structural VAR process:

(2) $$B_0 R_t = a + \sum_{k=1}^{K} B_k R_{t-k} + \eta_t,$$

where $\eta_t = [\eta_{N,t}, \eta_{D,t}]'$ is the vector of structural shocks and $a=[a_N, a_D]'$ is the vector of intercepts. The shocks $\eta_{N,t}$ and $\eta_{D,t}$ are serially uncorrelated and reflect respectively the overnight and daytime changes in economic fundamentals, microstructure factors, or short-term mispricing. Their variances are normalized to one. Therefore $E(\eta_t)=0$; $E(\eta_t \eta_{t-k}') = 0$ for $k \neq 0$; $E(\eta_t \eta_t') = I$, a 2×2 identity matrix. B_0 is a lower triangular matrix because the periods are sequential: within the same day t, $r_{N,t}$ affects $r_{D,t}$ but not vice versa. The impact of daytime trading on overnight returns is captured by the lagged returns on the right hand of (2).

The corresponding reduced form of the structural VAR is given by $A(L)R_t = \alpha + \varepsilon_t$, where $A(L) = I - A_1L - ... - A_KL^K$, $A_k = B_0^{-1}B_k$, and $\alpha = B_0^{-1}a$. The vector of reduced-form shocks is given by $\varepsilon_t = B_0^{-1}\eta_t$. Let $\iota = [1,1]'$ and $h' \equiv [h_N, h_D] = \iota' A(1)^{-1} B_0^{-1}$. Using the Beverage-Nelson decomposition, Wang and Yang (2011) show that the parameters h_N and h_D reflect the impact of the shocks $\eta_{N,t}$ and $\eta_{D,t}$ on the changes in the efficient price, in this case the true value of the KRW/USD rate. The overnight and daytime information shares are

(3) $$IS_i = \frac{h_i^2}{h_N^2 + h_D^2}, \; i = N \text{ or } D.$$

Note that A(1) in the reduced-form VAR is easily estimated by OLS and the B_0^{-1} matrix is the lower triangle Cholesky factor of the estimated variance matrix of ε_t. Hence the IS requires the same data as the WPC and is almost as easy to compute.

3. Comparison between WPC and IS

Wang and Yang (2011) show that the $IS_N = \frac{var(\Delta m_{N,t})}{var(\Delta m_t)}$ where m_t is the unobservable efficient price on day t. Wang and Yang (2012) show when returns are normally distributed

Electronic copy available at: https://ssrn.com/abstract=1972641

with mean zero, $WPC_N \rightarrow \frac{\sigma_N^2+\rho_{ND}\sigma_N\sigma_D}{\sigma^2}$ in large sample where $\sigma_N^2 = var(r_{N,t})$, $\sigma_D^2 = var(r_{D,t})$, $\sigma^2 = var(r_t)$, and $\rho_{ND} = cor(r_{N,t}, r_{D,t})$. This highlights the difference between the IS_N and the WPC_N. The IS_N is based on the variation in the efficient price, the unobservable true value of the asset. The WPC_N is the variance ratio, adjusted for correlation, of the observed return. As the observed returns are often driven by microstructure noise and other market frictions, the IS is the more appropriate measure of price discovery. Wang and Yang (2012) show that the WPC converges to the IS when the returns follow independent normal distributions with zero mean, and it diverges from the IS when overnight and daytime returns are correlated or the overnight and daytime variance ratio is high. This paper uses both the WPC and the IS to measure price discovery in KRW. One should keep these properties in mind when interpreting the empirical findings.

III. Empirical Findings

1. Data and Summary Statistics

Before 1998, KRW had a managed float regime with a daily trading band. The band increased from 0.4% to 2.25% during 1990 to 1995. As the Asian financial crisis spread from Thailand to Korea, the band was widened to 10% on November 19, 1997 and was abolished on December 12, 1997.[1] Because the band was enforced in the onshore market, it restricts the information flow into the KRW rate during the Korean business hours. Therefore the estimated price discovery measures tend to be biased against the onshore market. To avoid the issues related to the trading band and the Asian financial crisis, I choose the sample period from January 1, 1999 to December 31, 2012.

The Bank of Korea publishes daily open, high, low and close KRW/USD exchange rates during the Korean business hours. For each trading day, the overnight and daytime returns are calculated as section II. Daily volatility is the natural log of the daily high-low

[1] See Chung, et al. (2000) for details of the KRW exchange rate regimes.

Electronic copy available at: https://ssrn.com/abstract=1972641

ratio. For each year I calculate the accumulative overnight and daytime returns as well as overnight and daytime return volatilities. Note that because the exchange rate is quoted as the number of KRW per USD, a *positive return* indicates a *depreciation* of KRW relative to USD.

Figure 1 shows the KRW/USD daily exchange rate and volatility over the sample period. Table 2 presents some summary statistics for overnight and daytime returns. There is strong evidence that KRW tends to depreciate overnight and appreciate during the Korean business hours. The accumulative overnight return Σr_N is positive, i.e. KRW depreciates overnight, in 10 of the 14-year sample. The accumulative daytime return Σr_D is negative, i.e. KRW appreciates during daytime, in 9 of the 14 years. The table shows that offshore markets have a very different view on KRW relative to domestic markets. The correlation between overnight and daytime returns ρ_{ND} is negative in every year and significantly negative in 9 of the 14 years.[2] In 1999 following the Asian financial crisis, KRW depreciated 22.6% in offshore markets but appreciated 28.3% in onshore markets, a difference over 50%! Domestic investors turned pessimistic during the global financial crisis in 2008: KRW depreciated 32.4% in onshore trading but appreciated slightly overnight. On average, KRW depreciated 3.89% per year overnight and appreciated 4.73% per year during daytime.

KRW volatility surged during the global financial crisis in 2008-09. Even though there is no offshore spot trading, the ratio of overnight to daytime return volatilities has been increasing steadily over the sample period. The average ratio is 86% over the sample period. In four of the last five years, overnight volatility is the same or higher than daytime volatility. The rising overnight-daytime volatility ratio provides the first indication that the overnight offshore news, events, and trading have an increasing impact on the KRW value, more than the impact of domestic onshore news and trading. [TABLE 2 ABOUT HERE]

[2] The significance of the correlation coefficient $\hat{\rho}$ is based on $t = \frac{\hat{\rho}}{\sqrt{(1-\hat{\rho}^2)/(N-2)}}$ where N is the sample size.

Electronic copy available at: https://ssrn.com/abstract=1972641

2. Overnight Price Discovery

The structural VAR model (2) provides the parameters for calculating the IS in (3). The corresponding reduced form VAR $A(L)R_t = \alpha + \varepsilon_t$ is estimated via OLS. The number of lags is determined by the Bayesian information criterion. The B_0^{-1} matrix is the lower triangle Cholesky factor of $\Omega = E(\hat{\varepsilon}_t \hat{\varepsilon}_t')$. The information shares in (3) are calculated from $h' = \iota' A(1)^{-1} B_0^{-1}$ where $\iota = [1, 1]'$. Wang and Yang (2011) provides a bootstrap procedure for assessing the statistical significance of the structural parameters. My interest here is not the statistical significance of individual parameters in the structural VAR model. I focus on the level of and trend in the overnight information share.

Table 3 reports the estimated overnight information share and the overnight weighted price contribution for each year in the sample. It is well known that the lag structure can be affected by data outliers. I estimate the IS and WPC with and without a filter which removes days with overnight or daytime return outside 5 standard deviations of the annual average. In most years, the 5 standard deviation filter removes one observation or less. The information shares, with and without filtering, are plotted in Figure 2 together with the WPC with filtering. There are several features in Table 3 and Figure 2:

- There is a strong upward trend in overnight price discovery in all measures over the sample period. The overnight information share rose from less than 10% in 1999-2000 to around 50% in 2011-2012. The overnight WPC rose from less than 20% in 1999-2000 to around 60% in 2011-2012. The rise is remarkable given there is no spot trading in overnight offshore markets. It is consistent with the rising offshore trading of KRW derivatives reported in Table 1 and reflects the aggregate effect from offshore holding and trading of KRW-related assets. The drops in the overnight information share in 2005 and 2009 coincide with the drops in the overnight information-to-noise ratio in these years, as discussed in section III(3).

Electronic copy available at: https://ssrn.com/abstract=1972641

- In most years, the estimated WPC_N are higher than the estimated IS_N. This is consistent with the theoretical and empirical findings of Wang and Yang (2012). Corollary 1 of Wang and Yang (2012) shows that the deviation between WPC_N and IS_N is large when ρ_{ND} is large (positive or negative). Large negative ρ_{ND} tends to increase $WPC_N - IS_N$, after controlling for other return characteristics (Wang and Yang, 2012, Table 5). As discussed earlier, the overnight offshore markets tend to have the opposite view on KRW relative to the domestic market, and ρ_{ND} is negative in every year in the sample period (Table 2). This explains the higher WPC_N relative to IS_N.
- Removing extreme returns have a strong effect on the lag structure of the structural VAR model. The Bayesian information criterion selects much shorter lags after filtering. However, Figure 2 shows that the estimated information shares are quite similar with and without filtering. [TABLE 3 AND FIGURE 2 ABOUT HERE]

3. Learning from Overnight Returns

If indeed the overnight price discovery has been increasing, then overnight returns should not be reversed during the Korean business hours. I use the unbiasedness regression of Biais, Hillion, and Spatt (1999) to test this projection. The daily (close-to-close) return r_t is regressed on the overnight (close-to-open) return:

$$r_t = \alpha + \beta r_{N,t} + \varepsilon_t \quad (4)$$

Biais, Hillion, and Spatt (1999, p1234) use the estimated β to gauge the information content of the opening price and the amount of learning by daytime traders: if $r_{N,t}$ is pure noise and is ignored by daytime traders, then $\beta = 0$; if $r_{N,t}$ is driven by information which is then absorbed by daytime traders, then $\beta = 1$. Barclay and Hendershott (2008) argue that the OLS estimator of β can be interpreted as the overnight information-to-noise ratio. I believe that overnight information is not entirely captured by β: If more information is incorporated into overnight price movements, one would expect that overnight volatility to be higher. Therefore the

Electronic copy available at: https://ssrn.com/abstract=1972641

overnight volatility has information content independent of the slope coefficient β. The impact of overnight offshore markets should be measured by the regression R^2, which is given by $\frac{\beta^2\sigma_N^2}{\sigma^2}$. Holding β constant, greater overnight price discovery increases overnight return variance σ_N^2 and the regression R^2.

Table 4 reports the regression results of equation (4), including the estimated β, its standard error, and the regression R^2 for each year. These results are also depicted in Figure 3. The standard error of the estimated β is based on the Newey-West robust covariance with automatic lag selection using Bartlett kernel. There are several features in Table 4 and Figure 3. The estimated β is close to one in several years. I fail to reject β=1 in 1999, 2003, 2006, 2007, and 2010. The overnight market was highly efficient in these years, according to the Barclay-Hendershott interpretation of β. The overnight market was less efficient (β<1) in other years, e.g. 2005 and 2009. It is interesting to observe that the lower β coincides with the lower overnight information shares in 2005 and 2009, even though the two measures have very different definitions. Standard error of β has been declining over time. The rising statistical significance of β also indicates the rising importance of the overnight market, independent of the size of the estimated β. Last but not least, the regression R^2 has increased from less than 20% in 1999 and 2000 to over 50% in 2011 and 2012. The pattern of R^2 fits remarkably well with the price discovery measures in Figure 2. Taken together, the evidence suggests that the observed daily changes in the KRW value are increasingly driven by changes in the overnight offshore markets. [TABLE 4 AND FIGURE 3 ABOUT HERE]

4. Weekend Price Discovery

The increasing price discovery in overnight offshore markets indicates greater internationalization of KRW. Conversely when there is no news in offshore markets, offshore price discovery should be very low and should not rise with currency internationalization.

Electronic copy available at: https://ssrn.com/abstract=1972641

Weekends generally have few business news and little trading activity. While Friday night in Korea is Friday daytime in the U.S., there is a long period from U.S. Friday close to Seoul Monday open during which businesses and most financial markets are closed. Therefore weekends should have lower information share than regular trading days. More importantly, despite the rising level of currency internationalization, one would not expect a significant rise in the weekend IS over time given the lack of relevant information. [TABLE 5 AND FIGURE 4 ABOUT HERE]

I estimated the weekday versus weekend IS_{WE} and WPC_{WE}. At weekly frequency, the 5-standard deviation filter does not filter out any return, therefore has not effect on IS_{WE} and WPC_{WE}. The weekly regression $r_w = \alpha + \beta r_{WE,w} + \varepsilon_w$ is estimated, where r_w is the weekly return based on Friday close KRW/USD rates and $r_{WE,w}$ is the weekend return from Friday close to Monday open. Table 5 and Figure 4 present the estimated weekend IS_{WE}, WPC_{WE}, and the regression results. The average IS_{WE} and WPC_{WE} are around 11%, approximately 4.25% per day in weekends. This is much lower than the average IS per weekday of 20.5%.[3] While there is considerable variation in IS_{WE} and WPC_{WE}, there is no significant trend in IS_{WE} and WPC_{WE} over the sample period. Compared to the daily regression reported in Table 4, the estimated β has much higher standard errors and greater variation. The R^2 closely resembles IS_{WE} except for 2006. Overall the evidence shows that (1) weekends have much lower information shares compared to weekdays, and (2) weekend IS has not increased over the sample period given there is relatively little information flow in offshore markets. Both findings provide indirect support for using IS as a measure for currency internationalization.

IV. Conclusion

This short paper proposes a new measure for the level of internationalization of a currency. Instead of using statistics on international trades or asset holding and trading, I

[3] Note that there are 2 and 2/3 days from Friday close to Monday open, assuming 1/3 day for daytime (8 hours) and 2/3 day for overnight. This leaves 4 and 1/3 days from Monday open to Friday close, with information share per day = (1-0.11)/4.33 = 20.5%.

Electronic copy available at: https://ssrn.com/abstract=1972641

propose to measure internationalization by the contribution to price discovery from offshore markets. Applied to data on the Korean won, I find that the overnight offshore markets play an increasingly important role in the price discovery of the KRW/USD exchange rate. Future research should explore how market conditions and regulatory changes affect the internationalization process.

Reference

Barclay, M. J., and Jerold B. Warner, 1993, Stealth trading and volatility: Which trades move prices? *Journal of Financial Economics* 34, 281-305.

Barclay, M., and T. Hendershott, 2003, "Price Discovery and Trading after Hours," *Review of Financial Studies* 16, 1041-1073.

Barclay, M., and T. Hendershott, 2004, "Liquidity externalities and adverse selection: evidence from trading after hours", *Journal of Finance* 59, 681–710.

Barclay, M., and T. Hendershott, 2008, "A comparison of trading and non-trading mechanisms for price discovery," *Journal of Empirical Finance* 15, 839-849.

Biais, B., P. Hillion, and C. Spatt, 1999, "Price discovery and learning during the preopening period in the Paris Bourse," *Journal of Political Economy* 107, 1218–1248.

Cao, C., E. Ghysels, and F. Hatheway, 2000, "Price discovery without trading: Evidence from the Nasdaq preopening," *Journal of Finance* 55, 1339-1365.

Chung, Chae-shick, Yang, Doo-yong and Joo, Sangyoung. 2000. Foreign Exchange Market Liberalization: The Case of Korea. Korea: Korea Institute for International Economic Policy.

Cliff, M., and M. J. Cooper, and H. Gulen, 2008, "Return Differences between Trading and Non-trading Hours: Like Night and Day," manuscript, Virginia Tech.

Ellul, A., Shin, H., Tonks, I., 2005, "Opening and closing the market: Evidence from the London Stock Exchange," *Journal of Financial Quantitative Analysis* 40, 779–801.

Hasbrouck, J., 1995, "One Security, Many Markets: Determining the Contributions to Price Discovery," *Journal of Finance*, 50(4), 1175-1199.

Huang, R. D., 2002, "The quality of ECN and Nasdaq market maker quotes," *Journal of Finance* 57, 1285-1319.

Kim, Kyungsoo, and Young Kyung Suh, 2009, "Dealing with the benefits and costs of internationalization of the Korean won." Paper presented at the BoK/BIS seminar on *Currency internationalisation: lessons from the global financial crisis and prospects for the future in Asia and the Pacific*, Seoul, Korea, 19–20 March 2009.

McCauley, Robert, 2006, "Internationalising a currency: the case of the Australian dollar," BIS Quarterly Review, December, 41-54.

Rhee, Gwang-Ju, 2009, "The recent experience of the Korean economy with currency internationalization."

Wang, J. and M. Yang, 2011, "Housewives of Tokyo versus the Gnomes of Zurich: Measuring Price Discovery in Sequential Markets," *Journal of Financial Markets* 14, 82-108.

Wang, J. and M. Yang, 2012, "How Well Does the Weighted Price Contribution Measure Price Discovery?" working paper, University of Technology Sydney.

Electronic copy available at: https://ssrn.com/abstract=1972641

Table 1: Onshore and Offshore Trading of KRW

This table summarizes the global trading of KRW in billion USD, including spot, forward, swaps, and options. It is based on the BIS triennial surveys of global foreign exchange market activity.

	Global Trading	Onshore Trading	Offshore Trading	Offshore Percentage
2001	9.76	8.42	1.34	14%
2004	21.15	17.14	4.01	19%
2007	34.05	27.11	6.94	20%
2010	60.27	37.92	22.35	37%

Table 2: Summary Statistics of Overnight and Daytime Returns

This table provides annual summary statistics for overnight and daytime returns. For a given year, Σr_N and Σr_D are the accumulative overnight and daytime returns respectively; ρ_{ND} is the correlation coefficient of overnight and daytime returns; σ_N and σ_D are the overnight and daytime volatilities respectively; σ is the volatility of daily (close-to-close) return; T is the number of observations. The asterisk * indicates significant at 5%.

	Σr_N (%)	Σr_D (%)	ρ_{ND}	σ_N (%)	σ_D (%)	σ_N/σ_D	σ (%)	T
1999	22.6	-28.3	-0.07	0.20	0.39	52%	0.42	251
2000	8.30	2.24	-0.16*	0.19	0.40	49%	0.41	244
2001	-1.53	5.33	-0.24*	0.37	0.42	90%	0.49	246
2002	1.70	-11.9	-0.22*	0.28	0.37	78%	0.41	245
2003	9.90	-9.36	-0.03	0.25	0.35	70%	0.42	248
2004	3.39	-17.6	-0.13*	0.29	0.30	98%	0.39	250
2005	0.94	-3.23	-0.23*	0.29	0.35	83%	0.40	249
2006	3.77	-12.2	-0.02	0.27	0.34	79%	0.43	247
2007	3.40	-2.73	-0.03	0.21	0.22	94%	0.30	246
2008	-2.76	32.4	-0.01	1.26	1.26	100%	1.77	248
2009	-6.11	-1.73	-0.15*	0.71	0.87	82%	1.04	253
2010	16.0	-18.6	-0.13*	0.62	0.60	104%	0.81	251
2011	-0.57	2.06	-0.13*	0.53	0.48	111%	0.67	248
2012	1.72	2.22	-0.11*	0.29	0.26	112%	0.36	248
Average:	3.89	-4.73	-0.12	0.41	0.47	86%	0.59	248
Median:	2.54	-2.98	-0.13	0.29	0.38	87%	0.42	248

Electronic copy available at: https://ssrn.com/abstract=1972641

Table 3: Overnight Information Share and WPC

WPC_N is the overnight weighted price contribution calculated from equation (1). IS_N is the overnight information share estimated from equations (2) and (3). SVAR Lags are the number of lags in the structural VAR model for IS. A 5-standard deviation filtering is applied when overnight or daytime return is outside 5 standard deviations of the annual average. ΔT is the change in the number of observations after filtering.

Year	No Filtering			ΔT	5 St. Dev. Filtering		
	IS_N (%)	WPC_N (%)	SVAR Lags		IS_N (%)	WPC_N (%)	SVAR Lags
1999	11.1	19.1	19	-2	5.8	19.8	8
2000	6.7	17.6	13	-1	6.7	16.1	9
2001	33.9	38.3	4	-1	29.9	36.0	2
2002	24.1	35.1	0	0	24.1	35.1	0
2003	30.8	35.8	0	0	30.8	35.8	0
2004	40.6	52.2	0	-1	42.9	53.3	0
2005	27.8	46.9	0	-1	24.4	45.7	0
2006	36.3	42.2	0	0	36.3	42.2	0
2007	45.6	50.1	0	0	45.6	50.1	0
2008	49.8	44.5	11	-3	41.0	44.8	0
2009	25.7	36.4	1	-1	26.1	35.1	0
2010	45.1	50.9	0	-1	49.0	48.9	0
2011	49.5	59.1	0	-1	53.8	60.7	0
2012	51.1	58.4	0	0	51.1	58.4	0
Average:	32.8	40.7	4	-1	33.4	41.6	1
Median:	33.9	42.2	0	-1	30.8	42.2	0

Electronic copy available at: https://ssrn.com/abstract=1972641

Table 4: Impact of Overnight Return

This table reports the estimates of the regression $r_t = \alpha + \beta r_{N,t} + \varepsilon_t$, where r_t is the daily close-to-close return and $r_{N,t}$ is the close-to-open (overnight) return. The standard error (St. Err.) is based on the Newey-West robust covariance with automatic lag selection using Bartlett kernel. The asterisk * indicates H_0: $\beta=1$ is rejected at 5% significance level.

Year	β	St. Err.	R^2
1999	0.96	0.151	0.18
2000	0.69*	0.178	0.09
2001	0.80*	0.068	0.31
2002	0.72*	0.089	0.25
2003	0.96	0.119	0.32
2004	0.87*	0.079	0.42
2005	0.70*	0.056	0.24
2006	0.98	0.087	0.37
2007	0.97	0.087	0.46
2008	0.78*	0.066	0.37
2009	0.76*	0.102	0.26
2010	0.97	0.068	0.50
2011	0.93	0.068	0.54
2012	0.90	0.054	0.51
Average	0.86	0.092	0.35
Median	0.88	0.082	0.34

Electronic copy available at: https://ssrn.com/abstract=1972641

Table 5: Weekend Information Flow

IS_{WE} is the weekend information share. WPC_{WE} is the weekend weighted price contribution. The regression $r_w = \alpha + \beta r_{WE,w} + \varepsilon_w$ is estimated, where r_w is the weekly return based on Friday close prices and $r_{WE,w}$ is the weekend return from Friday close to Monday open. The standard error (St. Err.) is based on the Newey-West robust covariance with automatic lag selection using Bartlett kernel.

Year	IS_{WE} (%)	WPC_{WE} (%)	β	St. Err.	R^2
1999	12.9	6.8	1.86	0.45	0.21
2000	5.9	4.2	0.59	0.75	0.01
2001	12.4	10.7	1.15	0.43	0.12
2002	13.3	11.0	0.95	0.32	0.11
2003	11.7	7.8	1.61	0.51	0.12
2004	15.8	16.3	1.06	0.34	0.14
2005	8.1	10.3	0.92	0.41	0.08
2006	16.0	13.0	0.61	0.31	0.05
2007	6.6	16.4	0.75	0.34	0.07
2008	13.0	8.5	1.65	0.78	0.14
2009	1.2	2.5	0.36	0.49	0.01
2010	13.8	12.9	0.89	0.39	0.14
2011	10.6	14.1	0.88	0.39	0.11
2012	15.1	20.4	0.75	0.24	0.13
Average:	11.2	11.1	1.00	0.44	0.10
Median:	12.4	10.7	0.91	0.40	0.12

Electronic copy available at: https://ssrn.com/abstract=1972641

Figure 1: KRW/USD Rate and Volatility

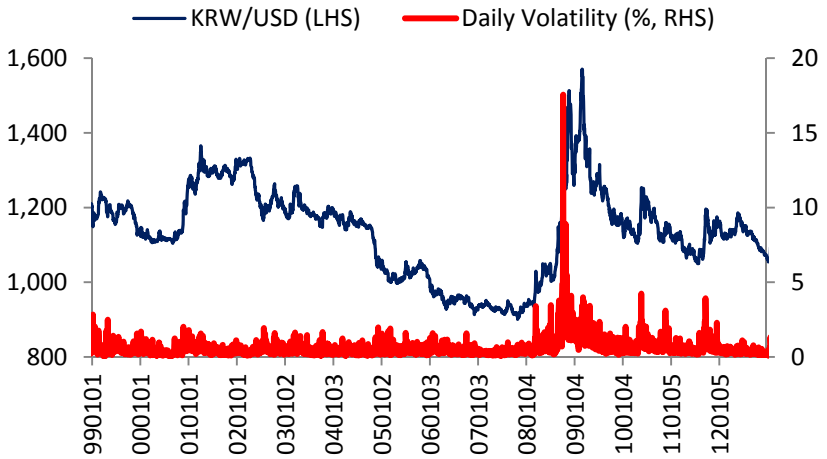


Figure 2: Overnight Price Discovery

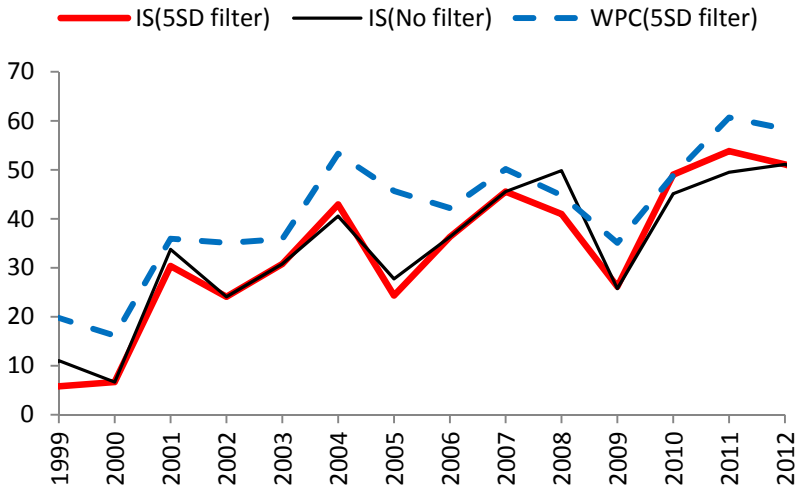
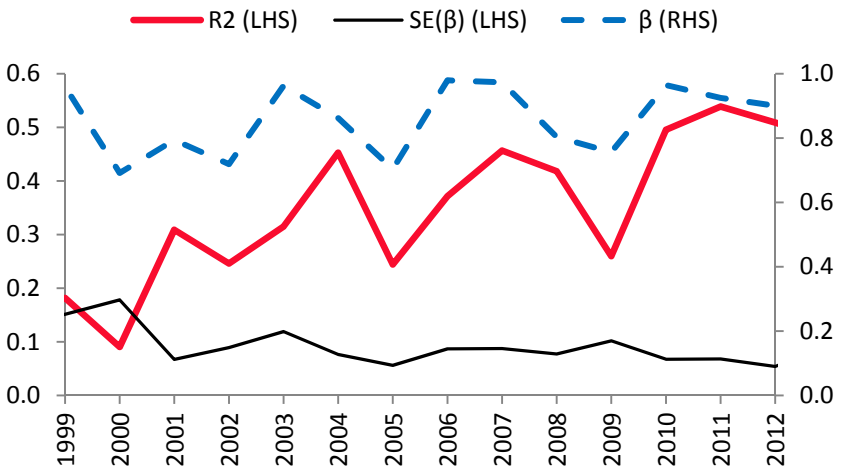
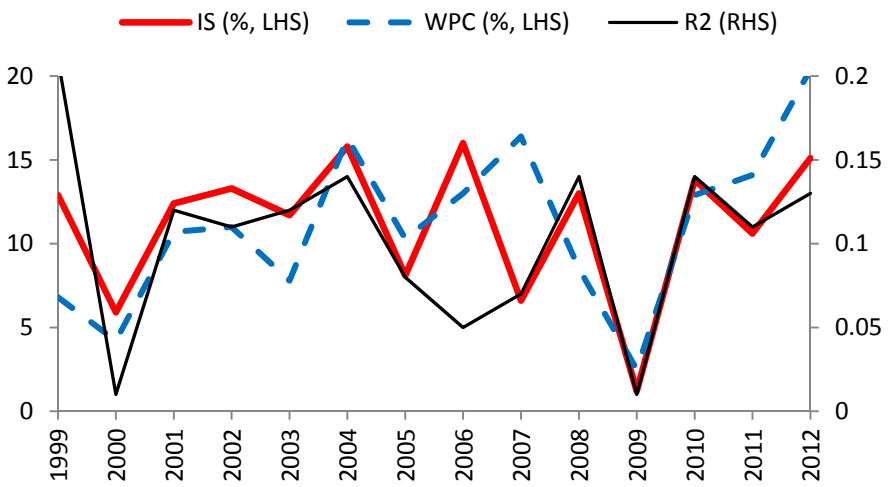


Figure 3: Impact of Overnight Return



Electronic copy available at: https://ssrn.com/abstract=1972641

Figure 4: Weekend Information Flow



Electronic copy available at: https://ssrn.com/abstract=1972641